COMP5424 Extended Reality

Prototype Design and Technical Note

Phase 2 – Group 7

Scenario D: Campus Emergency Preparedness

VR Stair-Evacuation and Lift-Safety Practice

Virtual client	Harbourview University Safety and Resilience Office
Submission date	17 October 2026
Repository	https://github.com/itemgroup/COMP5424_CampusEmergency_XR
Source baseline	final-submission-2026-10-17 at 9f2b3ff
Submitted build	Build/CampusEmergency.zip; Windows x86_64 IL2CPP, source 9f2b3ff
Target-PC evidence	Team-reported Windows PCVR test on PQ604B / 604B205; Quest Link/Air Link; 17 October 2026

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A supervised educational demonstration; not official emergency guidance, an approved campus evacuation plan, or certification.

1 Prototype Scope and Requirements Traceability

1.1 Purpose and implemented scope

Phase 1 proposed sign interpretation and recovery when an initial evacuation route becomes unavailable [1]. Phase 2 prioritises a runnable, testable XR journey rather than visual polish [2]. The current source implements **stair-evacuation and lift-safety practice** with an optional, authored route-closure and recovery exercise.

Implemented journey: Enter System → welcome and task briefing → stair route and fire alert → marked landings → 2F practice endpoint and persistent completion card. Optional reroute mode first presents the original route, then closes it at the 3F decision corner and requires a signposted same-floor alternative before rejoining the stairs. The diagram and controller tutorial remain optional. A separate G-floor courtyard assembly interaction is additional content, not the 2F route's completion condition.

1.2 Requirement-to-feature mapping

Status describes software scope, not demonstrated learning effectiveness. Evidence keys refer to Appendix A and the source baseline [3].

ID	Requirement / implementation	Status / evidence
R1	Complete evacuation practice journey: briefing, stairs, endpoint and optional recovery.	Implemented authored practice; learning effect not validated. S1–S3; V1, V3.
R2	Interpret signs and environmental cues: floor/lift signs, diagram and task card.	Implemented cues; independent interpretation not established. S1–S3; V2.
R3	Recover when an initial route becomes unavailable.	Optional 3F closure, enforced alternative and stair rejoin; authored, not dynamic routing. V3.
R4	Explain why an action or route is appropriate.	Lift warning and safety review explain stair/lift choice and closure cues; learning effect unmeasured. S2–S4; V3.
R5	Calm, inclusive, low-risk practice.	Partial: text feedback, optional tutorial, teleport and moderate audio settings; comfort needs user evidence. V4.
R6	Pause, exit and restart safely.	Implemented; PCVR pass reported for PQ604B/604B205; broader device coverage remains unverified. S2; V5, V6.

1.3 Changes from Phase 1

The route-change requirement is implemented as a bounded, optional exercise using one fixed closure, signs and an authored corridor, not as dynamic routing. Reusing the VR template and imported building assets enabled integration of movement, signs and feedback. The environment includes multiple floors, exceeding the proposed small scene. Campus geometry is illustrative, not a validated evacuation map. The prototype can be evaluated for task completion and cue usability, but current formative evidence cannot establish learning or transfer. AI co-navigation, multiplayer, certification and graphic emergency effects remain outside scope.

2 Interaction and Prototype Design

2.1 Core interaction and route decisions

The welcome NPC introduces the demonstration. Continue reveals the task list, evacuation diagram and optional five-step controller practice. Users then read corridor signs and enter the marked stair route. Arrival at its starting waypoint starts the exercise audio; ordered checkpoints update the route task state. Position-based triggers support both walking and teleport arrival.

Four floors contain eight lift signs. Nearby panels appear within 2.5 m horizontally and 1.25 m vertically of a sign. During the fire exercise, entering this area can announce that lifts must not be used; activating a panel adds warning text and an error cue. Panels sit 0.5 m above and 0.5 m along world $+Z$ from each sign to reduce surface intersection.

Decision and rationale: Fixed markers make arrival feedback repeatable and avoid dependence on NPC placement. The optional reroute exercise closes the initial 3F route, blocks invalid teleport destinations and cross-barrier shortcuts, then signposts an opposite same-floor corridor back to the existing stairs. It is an authored scenario, not dynamic route planning. Outside this exercise ordinary exploration is unchanged. The diagram's arrows and automatic lift announcement reveal directions, so they confound claims of unaided choice.

2.2 Locomotion, controls, and feedback

Input: The left stick moves; the right stick turns and its upward teleport input aims at a valid target. Release confirms teleport, the right grip cancels, and the right trigger interacts with supported objects/UI. Teleport reduces reliance on continuous travel; individual comfort has not been evaluated. The right stick uses 45-degree snap turning.

UI: Tutorial/navigation panels are world-fixed after opening. Nearer/Farther physically move a panel in 0.2 m steps (normal range 0.65–2.2 m); Review reopens the guide at the current viewpoint. Thus the panel does not chase head movement, although users can leave it behind. The completion card persists after arrival at the endpoint or End NPC, independently of Continue. It offers a safety review, a Continue button that leaves the current state unchanged, and a Back action that reloads the scene to clear the previous attempt and open the classroom welcome page. The review describes stairs, lift avoidance, noticing a closure and the simulated endpoint; it is not evidence of retained learning.

Pause/reset: Pause sets the simulation time scale to zero and opens a view-following menu; Resume restores it. Restart reloads the active scene; Exit quits the player. Input disabling while paused has not been separately verified.

2.3 Calmness, accessibility, and journey evidence

English text, symbols, high-contrast panels and adjustable panel distance provide visual support alongside audio. Low-intensity cues avoid deliberate flashes, smoke or graphic harm. Multilingual access, seated reach, silent-mode comprehension and individual comfort are not yet established.

3 Technical Implementation and Functional Verification

3.1 Architecture and major technical decisions

Configuration: Unity 6000.3.25f1; URP 17.3.0; XR Interaction Toolkit 3.5.1; Input System 1.20.0; OpenXR 1.17.1; Android IL2CPP/ARM64 for Quest 3 [3]; the Windows PCVR test is summarised below.

Control: ControllerGuidePanel tracks route start, stair entry, landings and completion from teleport-anchor positions; a separate 2.5 m check shows the End NPC card once without advancing tasks. Opt-in RouteChangeExercise adds the selection filter, closure barrier and alternative signs. NpcGuideDialogue manages dialogue cards; MainMenuController restores campus roots and reloads the classroom; CampusAudioDirector and LiftSafetyAudio manage audio.

Trade-offs: One scene preserves the XR rig; authored markers and offline speech reduce integration and network risk. Recovery is limited to one optional authored closure and corridor. The ordinary endpoint card does not require ordered completion; reroute practice does.

3.2 Executable and repository reproducibility

The Windows Standalone x86_64 IL2CPP build from 9f2b3ff is in Build/CampusEmergency.zip. The team reported launching it on PQ604B machine 604B205 on 17 October 2026 via Quest Link/Air Link, with Meta Quest Link as the active OpenXR Runtime; all intended features reportedly worked. No task-by-task result log was retained. This records one PCVR setup, not Android APK acceptance or compatibility with other PCs.

Handoff: Extract the ZIP and run My project.exe in CampusEmergency with Quest Link active and the data folder and player DLLs beside it.

3.3 Functional verification summary

C# compilation and **147 isolated Unity regression checks passed** for the current source, including XR teleport filtering, barrier collision, ordered recovery and two clean scene reloads. These are automated regressions, not participant trials or full-campus acceptance (Appendix A).

Key	Executed or required check	Result / evidence
V1	Endpoint before Continue; persistent card; task isolation.	Pass; ordinary mode can acknowledge arrival before ordered completion.
V2	Diagram, Continue, panel movement and limits.	Pass; exact labels in evidence file.
V3	Optional closure, barrier, valid recovery and ordered stairs.	Pass in isolated scene; authored scenario, not participant evidence.
V4	Lift proximity; muted journey.	Proximity passes; muted journey not recorded.
V5	Pause, clean classroom reset and exit.	Automated reset passes; PCVR test report is summarised in V6.
V6	PCVR build on lab PC.	Team-reported pass; 17 October 2026, PQ604B/604B205.

4 Limitations and Evaluation Priorities

4.1 Operational definitions: recovery and assistance

Phase 1 feedback requested explicit criteria for successful recovery and acceptable assistance. The following is a **provisional Phase 3 protocol**, not a measured result or client-approved safety standard. It becomes applicable only after an unavailable-route event is implemented.

Successful recovery: After the closure cue appears, the participant recognises the closure through observable behaviour or explanation, chooses the signposted alternative, and reaches the fictional endpoint without bypass. Pilot a 120-second active-time window from cue presentation to endpoint entry, excluding pauses and technical interruptions, then freeze the threshold before the main study. Record recognition, route selection and completion separately.

Assistance: H0 = no route-recovery hints; H1 = at most one non-directional prompt (e.g., "Look again at the signs"); H2 = repeated prompts, explicit directions or facilitator-led navigation. Log wording, source and time, including system prompts. Standard control briefing is separate from route help. The current arrow diagram/automatic directions must be recorded as directional assistance, not silently counted as H0.

Judgement: Report completed recoveries as H0 independent, H1 acceptably assisted, or H2 beyond the proposed allowance. Report timeouts, incomplete attempts, technical interruptions and comfort withdrawals separately, with denominators. Always provide safety/comfort help regardless of classification.

4.2 Evaluation priorities and evidence

E1 – Cue interpretation (R2): Observe missed signs, hesitation and control errors; ask which cue informed each choice. **E2 – Recovery/rationale (R1, R3, R4):** The closure, alternative and safety review are now implemented. Record event times and assistance; ask for rationale before the review and record post-review explanations separately. **E3 – Comfort/inclusion (R5, R6):** Observe reading/reach difficulty, pause/exit use and discomfort; participation is voluntary.

Study status: Group members provided iterative hands-on feedback, summarised in Appendix A. This is retrospective formative convenience testing, not an independent study. The number present at each iteration, individual attribution, exact test dates, device/build and standardised outcomes were not logged. GitHub commit dates bound the development period but are not treated as test timestamps. No success rate, time saving or learning gain is claimed. Future evaluation should record anonymous task outcomes, prompts and optional comments with consent; do not retain names, account IDs or headset recordings by default.

4.3 Known limitations and next priorities

1. Establish recovery for unfamiliar users (R1/R3/R4): The closure, alternative and review now exist, but there is no independent evidence that users notice the closure, choose the alternative or understand the rationale. **2. Establish delivery reliability (R6):** Repeat PCVR verification on additional required lab PCs or headset/runtime combinations. Test Android standalone deployment separately if it remains a delivery target. **3. Pilot clarity and assistance (R2/R5):** Check mute comprehension, turning text, fixed-panel reach and sign alignment on-device. Teammate familiarity, a single layout and absent transfer evidence limit conclusions.

References

[1] Group 7. *Requirements and Prototype Proposal: Scenario D — Campus Emergency Preparedness*. COMP5424 Extended Reality, 20 September 2026. Repository copy: GroupProjectDocuments/Group7-GroupComponent.pdf.

[2] COMP5424 Extended Reality. *Phase 2 assignment instructions: Purpose, Minimum Prototype Expectations and Submission Components*. Course instructions supplied by the team; publication date/version not provided. Includes the four-page technical-note and target-platform requirements.

[3] Group 7. *Campus Emergency XR prototype*, release branch final-submission-2026-10-17, source commit 9f2b3ff, 17 October 2026. https://github.com/itemgroup/COMP5424_CampusEmergency_XR. Implementation evidence: Packages/manifest.json, ProjectSettings/ProjectVersion.txt, ProjectSettings/ProjectSettings.asset, Docs/SoundImplementation.md, and Tools/Sound/Validation/SoundVerification.cs.

A Evidence Locations and Submission Checks

A.1 Source locators

Paths are relative to the repository root. This report describes final-submission-2026-10-17 at 9f2b3ff. The Windows PCVR ZIP is a separate build artifact produced from that source revision. The Android standalone APK was not rebuilt for this acceptance.

Key	Locator and interpretation
S1	Assets/Scenes/SampleScene.unity: Start/End NPCs, DownStair waypoints, LiftSign and startup menu.
S2	Assets/Scripts/UI/ControllerGuidePanel.cs: LateUpdate, AdvanceRoute, TryShowEndDialogue, AdjustPanelDistance, pause/restart/exit handlers.
S3	Assets/Scripts/Building/NpcGuideDialogue.cs: ShowIntroduction, ShowTaskGuide, ShowCompletionDialogue.
S4	Assets/Scripts/Audio/CampusAudioDirector.cs: IsNearLiftSign, WarnAgainstLift, UpdateEvacuation; Assets/Scripts/Audio/LiftSafetyAudio.cs.

A.2 Executed regression evidence

The complete saved PASS record from the isolated, no-graphics Unity project is reproduced below. The persistent test definitions are in Tools/Sound/Validation/SoundVerification.cs; execute using Tools/Sound/Verify-Sound.ps1 with the Unity editor path. **Result: PASS; 147 named checks.** The suite uses an isolated fixture and does not verify the entire campus or Quest build.

- 22 audio assets import successfully
- NPC voice matches scene text
- Dynamic task-list markers match baked NPC speech
- All eight lift interaction panels use the actual sign positions on four floors
- Lift panels and colliders are hidden when the player is away
- Elevator NPC and End NPC locations do not count as lift sign proximity
- Actual lift marker is detected: LiftSign 0
- Actual lift marker is detected: LiftSign 1
- Actual lift marker is detected: LiftSign 2
- Actual lift marker is detected: LiftSign 3
- Actual lift marker is detected: LiftSign 4
- Actual lift marker is detected: LiftSign 5
- Actual lift marker is detected: LiftSign 6
- Actual lift marker is detected: LiftSign 7
- Lift detection rejects other floors and positions outside 2.5 meters
- Offset anchor resolves to actual destination
- Training route uses corrected anchor position
- Evacuation diagram imports as Texture2D
- Start NPC welcome precedes the optional controller tutorial
- Task updates do not overwrite welcome before Continue
- Continue is an enabled XR-raycastable button
- Continue reveals controller tutorial and dismisses welcome button
- Task card includes the correct evacuation diagram with preserved aspect ratio
- Task list and teleport-effect instructions appear together
- Repeated Continue and bubble styling preserve the task card

- Guide Nearer physically moves its panel 0.2 meters without changing height or orientation
- Guide Farther physically reverses the Nearer adjustment
- Guide Nearer stops at 0.65 meters and disables its button
- Guide Farther stops at 2.2 meters and disables its button
- Guide remains world-fixed after head rotation and player translation
- Completing all five controller exercises opens the navigation panel
- Post-tutorial navigation panel does not follow head rotation, walking or teleport displacement
- Nearer adjusts a distant fixed HUD by one step instead of snapping it to the player
- HUD Nearer physically moves its panel while preserving height and orientation
- HUD Farther physically reverses the Nearer adjustment
- HUD Nearer clamps to 0.65 meters
- HUD Farther clamps to 2.2 meters
- Review explicitly reopens the fixed guide in front of the current viewer
- Skipping the guide also opens a world-fixed navigation panel
- Endpoint dialogue does not trigger outside actual anchor radius
- Endpoint dialogue does not trigger from another floor
- Inactive endpoint NPC does not consume arrival trigger
- Direct endpoint arrival shows configured End NPC text without prior route checkpoints
- Endpoint dialogue leaves ordered task progress unchanged
- Repeated endpoint arrival preserves the persistent card without replaying dialogue
- Completion card remains visible beyond its former seven-second timeout
- Completion card offers an optional controller-ray safety review button
- Completion card lower edge is raised 0.4 meters above the original panel lower edge
- Safety review explains stairs, lifts, closure awareness and the simulated endpoint without claiming reroute success
- Opening safety review preserves persistent display, voice cooldown and ordered route progress
- Taller safety review keeps the same raised lower edge, clear of the NPC head
- Raised review stays below the next floor at the End NPC's configured four-meter storey height
- Safety review body fits its panel at the configured minimum readable font size
- Safety review remains visible beyond the former dialogue timeout
- Back restores the configured completion message without replaying completion voice
- Reset after safety review clears the review page and persistent message for the next experience
- End NPC proximity does not trigger completion from another floor
- Approaching End NPC after reset restores the completion card rather than the previous review page
- Actual frame update still blocks endpoint completion while paused
- Actual frame update shows permanent completion before Continue at (48.72, 12.08, -10.44)
- Actual frame update shows permanent completion before Continue at (55.22, 12.08, -10.44)
- Completion before Continue plays or queues the configured completion voice
- Completion does not bypass Continue for the tutorial or advance ordered route progress
- Emergency starts idempotently
- Indoor alarm fades in to configured level
- Exact emergency PA is selected
- Speech ducks alarm for intelligibility
- Triggered NPC preserves currently displayed task-list text
- Triggered NPC queues the voice for the visible text
- XRI accepts real teleport request
- No teleport sound on request/aim/cancel phase
- Real XRI teleport completes and moves player
- Arrival audio is triggered by completed teleport
- Grounded continuous movement triggers footsteps
- NPC proximity cannot show the lift warning even when explicitly requested

- Lift proximity detected at matching floor
- Nearby lift marker enables its panel and XR collider
- Wrong lift warning is anchored at the matching lift marker, not the player or NPC
- Moving away hides lift panels and disables raycast targets
- Pause freezes alarm fades
- G-floor covered corridor does not count as outdoors
- G-floor open courtyard fades alarm without false completion
- Assembly point triggers success once
- Assembly retains only distant alarm
- EndEmergency clears alarm and speech queue
- Reroute validation is registered on the real XR Interaction Manager
- Optional reroute starts idle with no barrier or restrictions on ordinary exploration
- Original route sign text has viewer-correct left/right and up/down orientation from all four sides
- Start NPC task panel retains its original safe height below the next floor
- Start reroute button moves to the right panel edge, away from the NPC head
- Task card content restores its original unraised layout
- Optional exercise offers Start reroute while preserving the ordinary task diagram
- Starting reroute establishes the original route before any closure appears
- Repeated start cannot reset an active reroute exercise
- Exercise permits the initial corridor but explains why direct teleport to stairs is unavailable
- Visiting the alternative corner before the closure does not count as recovery
- Pause prevents a route closure from triggering
- Standing below the decision corner cannot trigger closure from another floor
- Reaching the decision corner changes the previously available route to closed exactly once
- Remaining at the closure does not repeat the event
- The visible closure has a static solid collider across the corridor, without a falling Rigidbody
- A real CharacterController continuous move is stopped by the closure instead of passing through
- Both fixed and arbitrary destinations beyond the barrier, downstairs and across the courtyard are rejected
- The registered selection filter rejects a real stair anchor before recovery
- A real XR ray hitting a free-area floor beyond the barrier cannot select it through the manager
- Valid corner-to-corridor teleports are allowed without treating their flight line as walking, while courtyard landings stay blocked
- A player near the barrier can teleport back to safety, but not through to the other side
- Notification regression uses an actively held real XR select input
- An actual invalid teleport gesture reports its reason once
- Holding the same invalid teleport gesture does not repeat a popup after the former four-second cooldown
- Releasing the teleport gesture rearms feedback for a new attempt
- A new invalid teleport gesture can report its reason again
- The same real XR ray and free-area floor remain selectable on the available alternative corridor
- The opposite same-floor corridor remains available for teleporting
- Alternative corner opens the connecting corridor, not an immediate success
- Rejoining stairs restores normal destinations but keeps the closed barrier non-teleportable
- Active exercise hides the answer diagram and start button, retaining environmental-sign guidance
- Recovery alone cannot show success before the ordered stair checkpoints are complete
- An out-of-order endpoint after recovery still cannot complete the stair route
- The original five stair checkpoints still complete in order after reroute recovery
- The real final arrival frame advances the last checkpoint before checking success, without a false incomplete warning
- Recovery plus completed stair checkpoints enables the persistent endpoint completion and review

- The final End NPC card offers Continue and Back alongside the retained safety review
- Continue does not move, rotate, reset progress, hide the card or replay completion voice
- Safety review uses Back to results and hides the classroom-return controls to avoid ambiguous Back buttons
- Returning from safety review restores both finish actions
- Dialogue reset hides both completion action buttons
- Disabling the exercise removes its global interaction filter
- Scene cleanup releases singleton and listeners
- Ordinary scene entry still displays Enter System and plays menu music
- The reloaded classroom reconnects all four tutorial input actions
- A fresh reroute can begin in clean attempt 0
- Restart fixture contains an active barrier, fire audio, popup and moved prop before Back
- Back unloads the previous experience including runtime objects, old XR rig and moved props
- Back skips Enter System and its music, opening the fresh classroom welcome instead
- Reload restores the classroom spawn and original prop placement
- Reload clears closure, task checkmarks, endpoint success and prior completion UI
- Reload resets fire/global state and time while registering exactly one new route filter
- The reloaded classroom reconnects all four tutorial input actions
- A fresh reroute can begin in clean attempt 1
- Restart fixture contains an active barrier, fire audio, popup and moved prop before Back
- Back unloads the previous experience including runtime objects, old XR rig and moved props
- Back skips Enter System and its music, opening the fresh classroom welcome instead
- Reload restores the classroom spawn and original prop placement
- Reload clears closure, task checkmarks, endpoint success and prior completion UI
- Reload resets fire/global state and time while registering exactly one new route filter
- Classroom skip is consumed once: the normal Restart action still opens Enter System

A.3 Retrospective group-member formative test log

Group members report hands-on iterations covering the full route, optional reroute, lift-safety prompts, controller/UI use and the Back reset. The issue notes below are consolidated from development feedback, not a contemporaneous test protocol. Individual attribution, per-session headcount, exact test dates and per-person task outcomes for earlier iterations were not retained. The Windows PCVR acceptance is recorded under V6. The GitHub development history spans 12–17 October 2026; this is not a claim that each iteration occurred on those commit dates.

Task / observation	Feedback recorded	Response and evidence
Controls and UI	Navigation/tutorial panels were reported to follow the player or distance controls appeared ineffective.	Panels are world-fixed; Nearer/Farther move them in bounded 0.2 m steps. Automated checks cover transform stability and distance limits.
Lift safety and completion	Lift warning appeared near the NPC rather than the lift signs; completion feedback was also reported missing or misplaced.	Lift prompts use the eight actual LiftSign markers. Completion is triggered by the final anchor or End NPC and remains visible. Regression checks cover marker proximity and endpoint behaviour.
Safety review	The Review safety button was obscured by the NPC head.	The shared completion/review card layout was raised; panel-bound checks pass in the isolated regression suite.
Optional reroute	The first route sign appeared mirrored and invalid-teleport notices recurred frequently.	Sign orientation was corrected; teleport validation now checks destination and barrier crossing, and rate-limits feedback per held gesture. Regression checks cover four viewing directions and repeated selection.
Back reset	Returning to spawn left traces of the previous attempt.	Back now reloads the scene and opens the classroom welcome directly. The automated suite verifies two consecutive clean reloads.
Windows PCVR build	The team reported successful launch and operation of all intended features on PQ604B/604B205 via Quest Link/Air Link; no itemised task log was retained.	Build/CampusEmergency.zip, source 9f2b3ff, 17 October 2026; single-machine PCVR result, not Android APK acceptance.

A.4 Interpretation of evidence

Automated regressions document implemented behaviour; the reported manual result documents one Windows PCVR setup. Neither constitutes a controlled learner study. No claims are made about learning gains, independent route choice, comfort, or compatibility beyond the tested PC/headset configuration.

A.5 Weekly Activity Record and Division of Work

The project progressed from the Unity/XR foundation through environment and interaction development to final integration and validation:

- **29 September–3 October:** Established the Unity/XR project structure, configured Quest/OpenXR settings, assembled the initial campus scene and began teleport, controller-turning and tutorial interactions (a1888f7, 79e93e7, 6c2beed, c54852e, d6cbf34, 4bb475e,

a962ae6, 49bddb1; PR #9).
- **5–9 October:** Added campus furniture, evacuation signs and posters and imported asset packages (384d062, 1aeddf4, 703edf0, b9ee901; PR #13).
- **12–16 October:** Refined the scene and UI and integrated lift-blocking behaviour, evacuation audio, a startup menu and English campus-resource descriptions (2ff3ece, d12625f, e0b82b9, 5693305, 786c9b6, 23e006f, 89d2419; PRs #17–#18).
- **17 October:** Integrated the optional reroute exercise, teleport-destination restrictions and clean Back/reset flow, then adjusted the Start reroute control. The project records 147 isolated Unity regression checks and a team-reported Windows PCVR test on PQ604B/604B205 via Quest Link/Air Link (416ca44, 364e4a6, 74234cb; PR #22).

Member	Responsibilities	Project evidence
SUN Jiashi	XR implementation and integration: campus scene, locomotion, teleport and turning, interaction flow, reroute logic, reset behaviour and final integration.	d6cbf34, f15de48, 364e4a6, 416ca44, 74234cb; changes across Assets/Scenes and Assets/Scripts. Phase 1 Development Group.
FENG Kedi	Unity development and interaction/scene work, including platform integration and the initial controller tutorial panel.	Phase 1 Development Group; feat/FKD–develop (PR #7) and feat/Cody-Scripts (PR #9, initial tutorial interaction panel).
SONG Zheyi	Campus resource images, signs and posters, English descriptions, evacuation audio and sound feedback; product/content coordination.	89d2419, 786c9b6; PRs #13 and #18. Phase 1 Product Group.
WENG Zhirui	Emergency-related content and asset preparation/handoff; product requirements and scope coordination.	a962ae6 (Module 5 content and asset handoff pack). Phase 1 Product Group.
ZHU Yu	Testing and acceptance responsibilities, together with Unity asset-package preparation.	Phase 1 Test Group; b9ee901 (Unity Asset Store packages and Git LFS).
CHU Silu	Test planning, acceptance and usability feedback.	Phase 1 Test Group; the Phase 2 feedback record covers UI, lift prompts, reroute signs and reset behaviour.

Prototype outcome: The delivered journey includes a classroom welcome and task briefing, a stair-based practice route to the 2F endpoint and an optional authored reroute exercise at the 3F decision area. The reroute is a bounded scenario, not dynamic route planning. The PCVR test covers one reported setup; Android standalone acceptance and structured participant evaluation remain future work.